

1. A company is building a web-based AI application by using Amazon SageMaker. The application will provide the following capabilities and features: ML experimentation, training, a central model registry, model deployment, and model monitoring. The application must ensure secure and isolated use of training data during the ML lifecycle. The training data is stored in Amazon S3. The company needs to use the central model registry to manage different versions of models in the application. Which action will meet this requirement with the LEAST operational overhead?

- a. Create a separate Amazon Elastic Container Registry (Amazon ECR) repository for each model.
- b. Use Amazon Elastic Container Registry (Amazon ECR) and unique tags for each model version.
- c. Use the SageMaker Model Registry and model groups to catalog the models.
- d. Use the SageMaker Model Registry and unique tags for each model version.

Answer C

2. An e-commerce company uses a Lambda function triggered by Amazon SNS to preprocess customer transaction data before sending it to an Amazon SageMaker model for fraud detection. The transaction data is then stored in Amazon RDS for long-term analytics. The company wants to ensure that only valid transactions (based on a predefined schema) are passed to the SageMaker model and that invalid transactions are logged and stored in Amazon S3 for further analysis. Which of the following solutions BEST fits the company's requirements?

- a. Use Amazon Kinesis Data Firehose to preprocess and validate the data, then trigger Lambda to send valid data to SageMaker, while invalid data is redirected to S3.
- b. Use Lambda to validate the schema, send valid data to SageMaker, and forward invalid data to an Amazon SQS queue, which will trigger another Lambda function to log and store it in S3.
- c. Use Lambda to validate the schema of each transaction, send valid data to SageMaker for inference, and write invalid data directly to Amazon S3.
- d. Set up a Step Functions workflow to manage validation and invoke Lambda for processing valid data and storing invalid data in S3.

Answer B - The approach provides a flexible and scalable solution for handling both valid and invalid transactions.

3. A company is building a web-based AI application by using Amazon SageMaker. The application will provide the following capabilities and features: ML experimentation, training, a central model registry, model deployment, and model monitoring. The application must ensure secure and isolated use of training data during the ML lifecycle. The training data is stored in Amazon S3. The company must implement a manual approval-based workflow to ensure that only approved models can be deployed to production endpoints. Which solution will meet this requirement?

- a.